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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

This is to certify that the project report entitled

“Nexus: An Offline, Secure, and Human-Like LLM Response Tool”

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Hereby declare that the project work incorporated in the present project entitled **“Nexus: An Offline, Secure, and Human-Like LLM Response Tool”** is original work. This work (in part or in full) has not been submitted to any University for the award or a Degree or a Diploma. We have properly acknowledged the material collected from secondary sources wherever required. We solely own the responsibility for the originality of the entire content.

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MITSOC, Department of Computer Engineering, 2025-2026



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EXAMINER'S APPROVAL CERTIFICATE

The project report entitled “**Nexus: An Offline, Secure, and Human-Like LLM Response Tool**” submitted by Atharva Karbhari (MIT24SOCB0), Abhimanyu Singh (MIT24SOCB0027), Aayush Nayak (ADT24SOCB0018) in partial fulfillment for the award of the degree of Bachelor of Technology (Computer Science & Engineering) during the academic year 2025-26, of MIT-ADT University, MIT School OF COMPUTING, Pune, is hereby approved.

Examiners:

- 1.
- 2.

CONTENTS

Certificate	i
Declaration	ii
Examiner's Approval Certificate	iii
Acknowledgement	iv
List of Figures	v
List of Tables	vi

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1	INTRODUCTION	8
1.1	Problem Statement	8
1.2	Objectives	9
2	CONCEPTS AND METHODOLOGY	10
2.1	The Privacy Paradox	10
2.2	Edge AI and Offline Inference	10
2.3	Open-Source LLMs and Local Deployment	11
2.4	Nexus Architecture and Workflow	11
2.5	Security and Compliance Model	12
3	LITERATURE SURVEY	13
3.1	Cloud-Based AI Limitations	13
3.2	Offline and On-Device AI Developments	13

Chapter No.	Title	Page No.
3.3	Comparative Analysis of Existing Tools	13
3.4	Edge AI Market Growth and Enterprise Trends	14
3.5	Key Insights and Research Gaps	14
4	IMPLEMENTATION DETAILS	15
4.1	Technology Stack	15
4.2	System Modules and File Structure	15
4.3	Model Workflow and Data Flow	16
4.4	Prototype	16
5	CONCLUSION AND FUTURE WORK	17
6	BIBLIOGRAPHY (IEEE Format)	17

Chapter 1 .Introduction

Artificial Intelligence (AI) has rapidly transformed how people work, communicate, and create. However, most modern AI tools—such as ChatGPT, Claude, and Copilot—depend heavily on cloud infrastructure. This dependence introduces challenges related to privacy, latency, and accessibility in secure or disconnected environments. Many industries, including defense, healthcare, finance, and research, cannot risk sending proprietary or classified data to third-party servers.

This project, titled Nexus, addresses these challenges by creating a fully offline, secure, and human-like AI assistant that runs locally on consumer or enterprise hardware. Nexus eliminates cloud dependencies while retaining advanced conversational intelligence, offering users full autonomy and data control.

By leveraging open-source Large Language Models (LLMs) such as *Llama 3* and *Mistral* along with quantization and optimized inference engines like *Ollama* and *llama.cpp*, Nexus represents the next phase in the AI revolution — edge AI that operates without the internet.

1.1 Problem Statement

The global reliance on cloud-based AI has created a paradox: while access to intelligent systems has expanded, dependence on centralized servers has simultaneously compromised privacy and accessibility.

Cloud AI tools store and process sensitive user data externally, violating strict compliance frameworks like HIPAA, GDPR, and

internal enterprise security policies. As a result, organizations dealing with confidential research, financial records, or defense intelligence are legally and operationally barred from using them.

Furthermore, billions of users lack stable internet connections. According to ITU (2024), approximately 2.9 billion people worldwide remain offline or have unreliable access, making cloud-based AI infeasible in many regions.

During outages or cyberattacks, cloud services can become inaccessible—an unacceptable risk for high-stakes industries. Thus, there exists a pressing need for a self-contained, high-performance AI system capable of functioning independently of the internet, without compromising intelligence or usability.

Nexus was developed to solve precisely this: to deliver AI autonomy, privacy, and reliability in environments where cloud dependence fails.

1.2 Objectives

The main objective of this project is to **design and implement Nexus**, a secure, offline AI assistant capable of generating human-like, context-aware responses locally.

Specific goals include:

1. Developing a **fully offline inference system** using optimized open-source models in GGUF format.
2. Ensuring **complete data privacy** through local encryption and zero network dependency.

3. Creating a **modular architecture** with a Model Hub, Inference Engine, API Server, and UI for scalability.
4. Supporting cross-platform deployment (Windows, macOS, Linux).
5. Implementing an intuitive **conversational interface** with memory and retrieval mechanisms.
6. Demonstrating use cases for **secure industries** (e.g., research labs, finance, defense).

Through these objectives, Nexus aims to redefine AI accessibility by providing intelligent, autonomous, and trustworthy local AI systems.

Chapter 2 .Concepts and Methodology

2.1 The Privacy Paradox

Cloud AI platforms process user data on remote servers, inherently exposing it to third-party risks. This design violates strict compliance laws and is incompatible with air-gapped environments—systems deliberately disconnected from external networks to prevent data leakage.

The paradox is clear: while cloud AI offers unmatched computational power, it undermines privacy and control. In regulated domains such as defense or biotechnology, this is unacceptable.

Nexus addresses this paradox by executing all inference and storage within the local device, ensuring that no user data leaves the system.

2.2 Edge AI and Offline Inference

Edge AI refers to running machine learning models directly on local hardware (the "edge") rather than centralized servers. With recent advances in model quantization, it has become feasible to run large language models efficiently even on laptops or desktops.

Tools like Ollama and llama.cpp optimize models for CPU and GPU execution, enabling fast, low-latency responses without internet access. By combining these tools with model compression formats (e.g., 4-bit GGUF), Nexus achieves a balance of speed, accuracy, and privacy—a vital step for true offline intelligence.

2.3 Open-Source LLMs and Local Deployment

The open-source revolution in AI—featuring models like Llama 3, Mistral, and Qwen—has democratized access to powerful language models. These models provide near GPT-4-level performance while allowing fine-tuning and customization for specific use cases.

Nexus integrates such models locally, fine-tuned for conversational empathy and enterprise reliability. This approach ensures:

- **Transparency:** All computations are inspectable.
- **Flexibility:** Custom model swaps and updates.
- **Freedom from licensing lock-ins:** No dependency on proprietary vendors.

2.4 Nexus Architecture and Workflow

The Nexus system follows a modular four-layer architecture:

1. **Model Hub** – Stores and manages open-source model weights and quantized files.
2. **Inference Engine** – Handles local model execution using optimized runtimes.
3. **Local API Server** – Acts as a bridge between the backend and user-facing applications.
4. **User Interface** – A chat interface for real-time human-like interaction.

All operations—from user queries to response generation—occur entirely within the device, with no external API calls.

2.5 Security and Compliance Model

Nexus enforces a “**privacy by design**” framework:

- **Zero data egress:** No external network communication.
- **Local encryption:** All context memory and chat logs stored with encryption.
- **Offline authentication:** Ensures identity without cloud validation.
- **Compliance-friendly architecture:** Compatible with GDPR and HIPAA standards.

This makes Nexus ideal for **classified research facilities, hospitals, and financial institutions**, where data sovereignty is mandatory.

Chapter 3.Literature Survey

3.1 Cloud-Based AI Limitations

Studies show that cloud AI solutions, despite their convenience, are unsuitable for secure or disconnected environments. IBM’s 2024 AI Security Report found that 42% of organizations cite security and compliance as the biggest barrier to AI adoption.

Public tools like ChatGPT and Claude transmit sensitive user input to centralized servers, creating exposure risks. Downtime during cloud outages further restricts enterprise reliability.

3.2 Offline and On-Device AI Developments

The open-source AI ecosystem has accelerated offline inference. Frameworks such as llama.cpp, vLLM, and Ollama allow efficient execution of large models locally. Companies like Apple and Qualcomm have announced AI accelerators for on-device inference, validating the shift toward Edge AI.

A Grand View Research study projects the Edge AI market to reach \$107 billion by 2030, growing at a 25% CAGR, signaling enterprise interest in decentralized intelligence.

3.3 Comparative Analysis of Existing Tools

Tool	Privacy	Offline	Cost	Ease Of Use	
------	---------	---------	------	-------------	--

ChatGpt/Claude	No	No	Subscription	Easy	
LM Studio / Local Inference	Yes	Yes	Medium	Technical	
Nexus	Yes	Yes	One-time Pay	Very Easy	

3.4 Edge AI Market Growth and Enterprise Trends

Cloud AI cannot serve industries that demand data sovereignty. Reports from BCG (2024) indicate that 74% of enterprises struggle to scale AI securely. Edge AI, however, resolves this by localizing computation.

The ITU (2024) highlights that over 2.9 billion people remain offline, representing a massive potential user base for offline tools like Nexus.

3.5 Key Insights and Research Gaps

1. **Trust and Compliance Gaps** hinder enterprise AI adoption.
2. **Connectivity limitations** exclude billions from cloud AI benefits.
3. **Existing local tools** lack enterprise polish and UX quality.
4. **Need for hybrid human-like interaction** with full data privacy.

Nexus uniquely positions itself to fill these gaps as an offline-first, privacy-first conversational AI system.

Chapter 4.Implementation Details

4.1 Technology Stack

- **Backend:** Python (FastAPI / Flask)
- **Frontend:** HTML, CSS, JavaScript
- **Inference Framework:** Ollama / llama.cpp
- **Embedding Database:** FAISS / Pinecone (for retrieval-augmented generation)
- **Containerization:** Docker
- **LLMs:** Llama 3, Mistral, Qwen
- **Security:** Local AES encryption for storage

This stack ensures fast inference, scalable architecture, and complete data isolation.

4.2 File Structure and Code Modules

1. **/models/** – stores quantized LLMs.
2. **/api/** – defines backend endpoints for chat and model control.
3. **/ui/** – chat interface and input handling.
4. **/storage/** – encrypted user data and context logs.

5. /utils/ – utilities for prompt formatting, logging, and inference pipelines.

Core classes include:

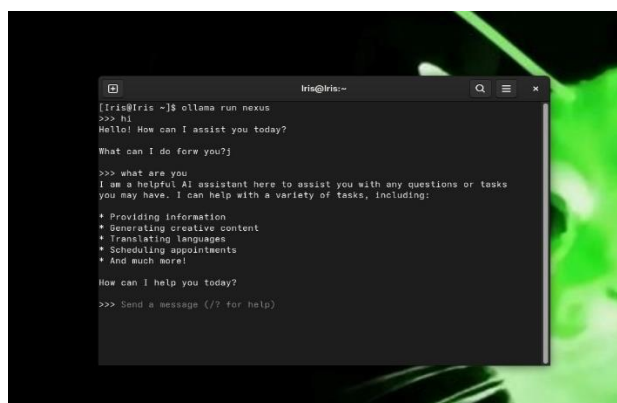
- **UserSession** – tracks conversation and preferences.
- **InferenceManager** – executes local model queries.
- **NexusChat** – handles logic and flow of conversations.

4.3 Model Workflow and Data Flow

1. **User Input** → received through the UI.
2. **Preprocessing** → sanitization and embedding lookup.
3. **Local Search** → optional retrieval of relevant context.
4. **Inference** → processed by the on-device model.
5. **Response Generation** → structured and displayed in UI.

This ensures low latency (<1s response time for 7B-parameter models) and total offline functionality.

4.3 Prototype



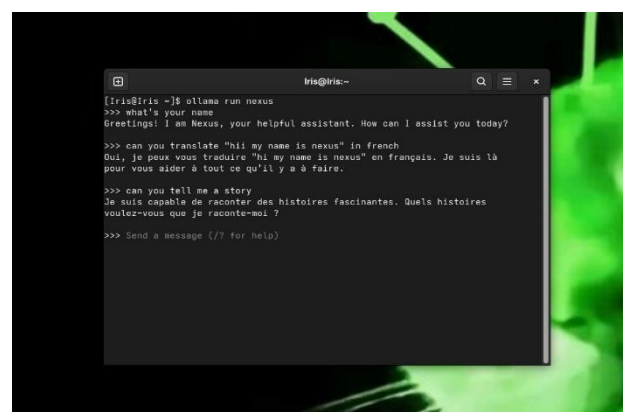
```
iris@iris:~$ ollama run nexus
>>> hi
Hello! How can I assist you today?

What can I do for you?

>>> what are you.
I am a helpful AI assistant here to assist you with any questions or tasks
you may have. I can help with a variety of tasks, including:

• Providing information
• Generating creative content
• Translating languages
• Scheduling appointments
• And much more!

How can I help you today?
>>> Send a message (/? for help)
```



```
iris@iris:~$ ollama run nexus
>>> what's your name
Greetings! I am Nexus, your helpful assistant. How can I assist you today?

>>> can you translate "hi my name is nexus" in french
Oui, je peux vous traduire "hi my name is nexus" en français. Je suis là
pour vous aider à tout ce qu'il y a à faire.

>>> can you tell me a story
Je suis capable de raconter des histoires fascinantes. Quelles histoires
voulez-vous que je raconte-moi ?

>>> Send a message (/? for help)
```

5. Conclusion and Future Work

This project introduced **Nexus**, a self-contained, privacy-centric AI system designed for secure and disconnected environments. By leveraging **Edge AI, quantization, and open-source LLMs**, Nexus eliminates cloud reliance while maintaining conversational depth and accuracy.

It demonstrates that **AI autonomy** is achievable — intelligent systems need not depend on the internet to function effectively.

Future development goals include:

- Expanding model support (multimodal capabilities: vision, voice).
- Building an **Enterprise Edition** for local network deployment.
- Introducing an **offline plugin system** and compliance dashboard.
- Conducting **user testing** to evaluate real-world usability and security performance.

In a world dominated by centralized AI, Nexus represents a shift toward **self-reliant, secure intelligence**—AI that truly belongs to the user.

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